CERTPATH PUBLISHING · FORMULA CHEAT SHEET

Journeyman Electrician — NEC Quick Reference

Based on NEC 2023 · verify against the latest adopted edition for your AHJ

⚡ CORE ELECTRICAL FORMULAS

Ohm's Law & Power

$V = IR \cdot P = VI = I^2R = V^2/R$

$kW = (V \times I \times pf) / 1000$ (1Ø)

$kW = (\sqrt{3} \times V \times I \times pf) / 1000$ (3Ø)

Voltage Drop — 2K Method

$VD = (2 \times K \times I \times L) / CM$ (1Ø)

$VD = (\sqrt{3} \times K \times I \times L) / CM$ (3Ø)

- K = 12.9 Cu · 21.2 Al (approx)
- L = one-way distance (ft).
- Target: **≤3% branch**, **≤5% total**.

📋 NEC LOOKUP TABLES

Ampacity — Table 310.16 @ 75°C Cu

AWG	Amps	AWG	Amps
14	20	1/0	150
12	25	2/0	175
10	35	3/0	200
8	50	4/0	230
6	65	250	255
4	85	500	380

Conduit Fill & OCPD Limits

# Conductors	Max Fill
1	53%
2	31%
Over 2	40%

OCPD 240.4(D): #14→15A · #12→20A · #10→30A

Box Fill — 314.16

- Each conductor = 1 count (include grounds & yokes).
- All grounds together = 1 count (largest).
- Yoke/device = 2 counts.
- Internal clamp(s) = 1 count (largest).
- Each conductor passing through = 1 count.

Motor Calculations

- Branch conductor: **125% of FLC** (430.22).
- Use **Table 430.250** for 3Ø FLC, not nameplate.
- OCPD: 250% non-time-delay, 175% time-delay (430.52).
- Disconnect within sight of controller.

🏠 SERVICES, GROUNDING, PROTECTION

Dwelling Service Load

- General lighting: **3 VA/ft²** (220.12)
- Small appliances: 2 × 1,500 VA (220.52)
- Laundry: 1,500 VA (220.52)
- Apply demand factors (220.42, .54, .55).

Grounding — 250.66

Service	GEC Cu
≤ 1/0	#6
2/0–3/0	#4

GFCI / AFCI

- **GFCI (210.8):** baths, kitchens (6 ft of sink), outdoors, garages, unfinished basements, laundry.
- **AFCI (210.12):** dwelling bedrooms, living rooms, kitchens, laundry, hallways, dens.

250–350 kcmil	#2
Rod/pipe/plate	#6 max (A)

📚 **Code citations to memorize:** 110.14(C) terminal temp · 210.8 GFCI · 210.12 AFCI · 220 load calcs · 240.4 conductor protection · 250 grounding · 310.16 ampacity · 314.16 box fill · 430 motors.